

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

Week-09

Progress and Integration Report

Project Title: AI Campus Assistant

Course Title: Mobile Computing Lab
Course Code: CSE 4204

Submitted by:

Team name: CSE4204-8C-T05
Section: 8C

Name of group members	IDs
Gazi Jayed Rasul	11220320906
Mirajur Rahman	11220320917
Md. Shakil Ahammed	11220320920
Sheikh Tashreeful Islam (team-leader)	11220321042

Submitted to:

Md. Riaz Mahmud
Assistant Professor
Department of Computer Science and Engineering

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Team Information

- **Team name:** CSE4204-8C-T05
- **Team Leader:** Sheikh Tashreeful Islam
- **Team Members:**

No.	Name	ID	Role
1	Gazi Jayed Rasul	11220320906	Database Manager
2	Mirajur Rahman	11220320917	Frontend Developer
3	Md. Shakil Ahammed	11220320920	AI Integration Lead
4	Sheikh Tashreeful Islam (team-leader)	11220321042	Backend Developer

- **Github Repository Link:** <https://github.com/TashreefulIslam/CSE4204-8C-T05-AI-Campus-Assistant>

1. Project Overview

AI Campus Assistant is an AI-powered web-based academic support system developed to help students, teachers, and administrators manage and access academic information more efficiently. The system provides role-based dashboards and academic management features along with AI-powered learning assistance.

The project has three main user roles: **Administrator, Teacher, and Student**. By Week 09, the major backend development and the main dashboards have been completed, and the team is currently working on finalizing the remaining Admin Dashboard modules and ensuring proper frontend-backend integration.

2. Summary of Completed Features

By Week 09, the team has completed the major foundation and core components of the project.

- **Backend:** The backend development has been completed. The backend provides the APIs and database communication required by the frontend application.
- **Database:** The database has been implemented and integrated with the backend for storing and managing the project's academic and user-related information.
- **Authentication:** The authentication system has been completed, including login, registration, logout, and role-based access for the three user roles.
- **Admin Dashboard:** The main Admin Dashboard has been completed. The Admin Dashboard provides administrative functionality for managing important academic and system information.
- **Teacher Dashboard:** The Teacher Dashboard has been completed. The teacher can access the teacher-related academic functionalities defined in the project, including assignment management, CT notices, course materials, student-related information, and AI quiz generation.
- **Student Dashboard:** The Student Dashboard has been completed. The student dashboard provides access to academic information and the AI-powered learning features defined in the project.
- **AI Features:** All planned AI features have been completed and integrated into the project. The AI functionalities include: AI Chat Assistant, AI Study Planner, AI Assignment Helper, AI Quiz Generator, AI Note Summarizer.

These features are available according to the appropriate user role and project workflow.

3. Partially Completed Features

At the current stage, two areas of the Admin Dashboard are partially completed.

3.1 Enrollment Management

The Enrollment Management module in the Admin Dashboard has been partially completed.

The basic functionality and interface have been developed, but some final integration and refinement work is still required.

3.2 Notice Management

The Notice Management module in the Admin Dashboard has also been partially completed.

The main interface and functionality have been developed, but the remaining integration and refinement work needs to be completed.

4. Remaining Features

The remaining development work is mainly focused on completing the two partially implemented Admin Dashboard modules.

4.1 Enrollment Management

The remaining work includes:

- Completing the frontend-backend integration.
- Testing enrollment operations.
- Checking database updates.
- Handling validation and errors.
- Final UI refinement.

4.2 Notice Management

The remaining work includes:

- Completing the frontend-backend integration.
- Testing notice operations.
- Checking database updates.
- Handling validation and errors.
- Final UI refinement.

After completing these two modules, the team will focus mainly on final testing, debugging, and UI/UX improvement.

5. Frontend – Backend Integration Status

The backend development has been completed, and the major frontend dashboards have also been completed.

The frontend is being connected with the backend through APIs so that the application can retrieve and update actual database information.

The integration has been completed for the major parts of the system, while the Enrollment Management and Notice Management modules in the Admin Dashboard require final integration and testing. The team is also checking:

- API requests and responses
- Database operations
- Form submissions
- Data retrieval
- Data updates
- Authentication
- Error handling

The main goal is to ensure that the final system uses actual backend data instead of temporary or hardcoded data.

6. AI Integration Status

All planned AI features of the project have been integrated successfully. The completed AI features are:

- ✓ **AI Chat Assistant** – provides academic question-answering support.
- ✓ **AI Study Planner** – generates personalized study plans.
- ✓ **AI Assignment Helper** – provides assistance with academic assignments.
- ✓ **AI Quiz Generator** – generates quiz questions based on user-provided topics or learning content.
- ✓ **AI Note Summarizer** – summarizes academic notes and study materials.

7. Authentication Status

The authentication system has been completed. The system supports three roles:

- ✓ Administrator
- ✓ Teacher
- ✓ Student

The authentication workflow includes:

- i. User registration
- ii. User login
- iii. Password handling
- iv. Role identification
- v. Role-based dashboard redirection
- vi. Protected access
- vii. Logout

After successful login, users are redirected to the dashboard according to their assigned role.

8. Major Problems Encountered

During the development process, the team faced several technical challenges.

8.1 JWT Authentication and PHP Version Problem

Initially, the team planned to use JWT authentication. However, the selected JWT package required **PHP 8.3 or higher**, while the project environment was running on **PHP 8.2**.

Because of this compatibility issue, the JWT package could not be properly used in the project.

8.2 Authentication Package Compatibility

The JWT package also created dependency and installation-related problems during backend development because of the PHP version and required extensions.

8.3 Frontend–Backend Integration Issues

While connecting the React frontend with the Laravel backend, some issues occurred with API responses, authentication data, and displaying backend data correctly in the dashboards.

These issues required repeated API testing and frontend debugging.

9. Solutions Implemented

- ✓ To solve the JWT and PHP compatibility problem, the team changed the authentication approach and used **Laravel Sanctum** instead of JWT.

Laravel Sanctum was selected because it works properly with the project's current Laravel and PHP environment and provides the required authentication functionality for the web application.

- ✓ For frontend-backend integration problems, the team tested the APIs using Postman and then connected the working API endpoints with the React frontend.

The team also debugged API responses and frontend components to ensure that the received backend data was displayed correctly.

- ✓ For the remaining Admin Dashboard issues, the team is currently completing the API integration and testing of the Enrollment and Notice Management modules.

10. Changes from Original Plan

No major changes were made to the original project plan.

The project has been developed according to the approved Project Proposal, SRS, System Design, and UI/UX plan.

The major features and overall system scope remain the same as originally planned.

Although the authentication implementation required a technical adjustment from JWT to Laravel Sanctum because of the project's PHP environment, this did not change the planned functionality of the system. The required authentication and role-based access functionality were implemented successfully.

11. Current Project Screenshots

The following screenshots should be included in the Week 09 report as required by the instructor.

Figure 1: Registration Page

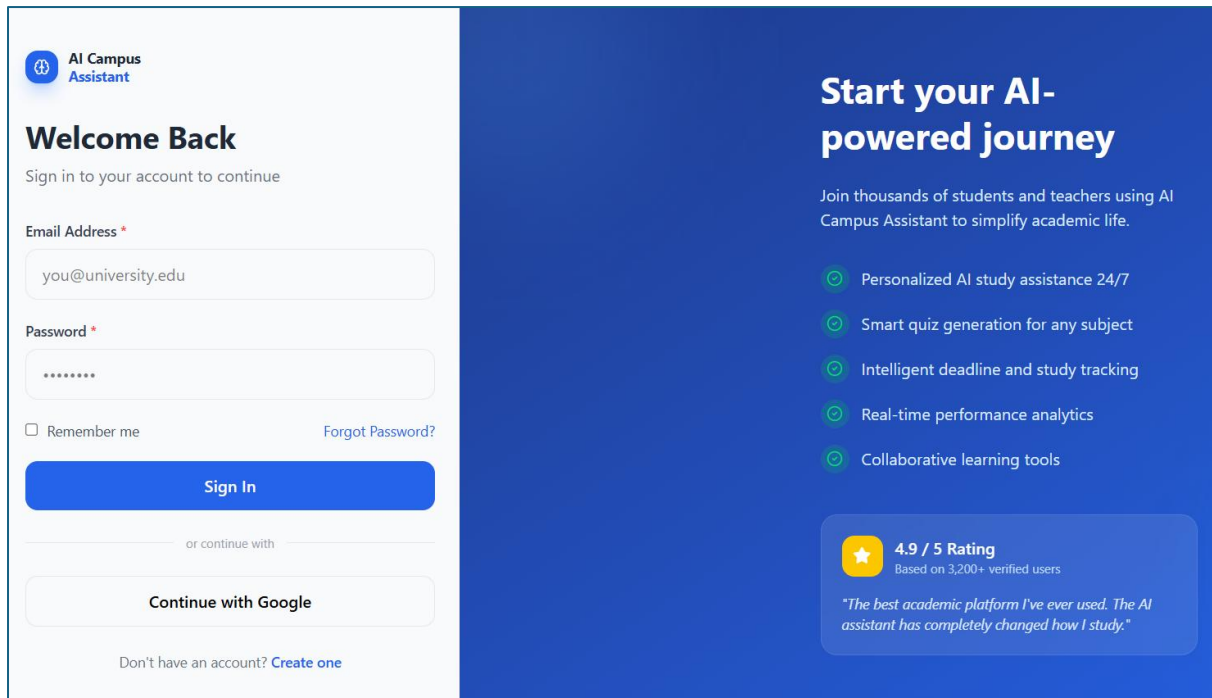


Figure 2: Login Page

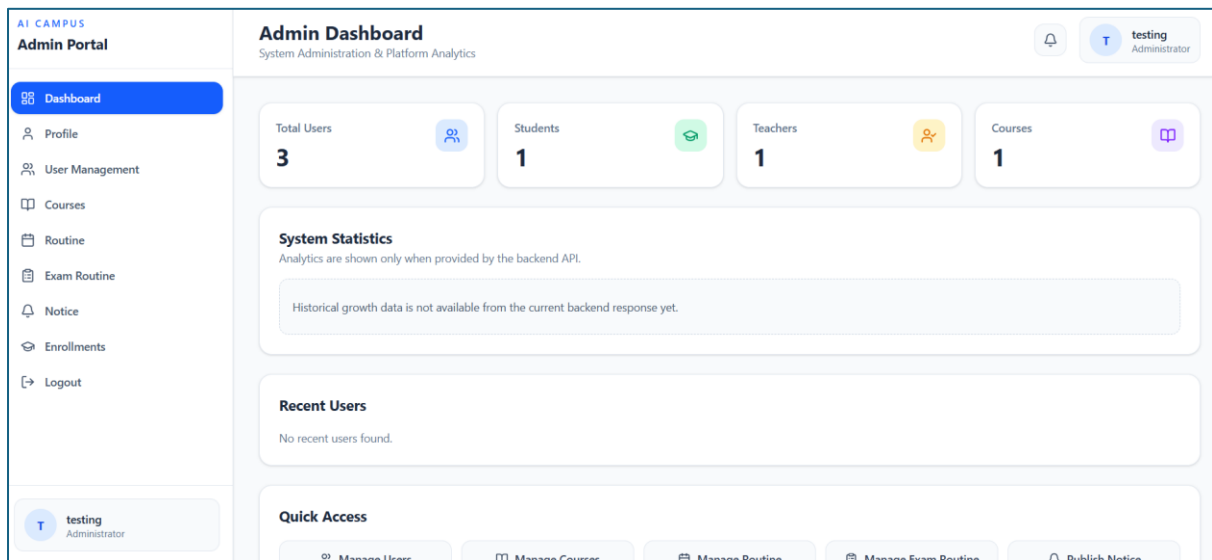


Figure 3: Admin Dashboard

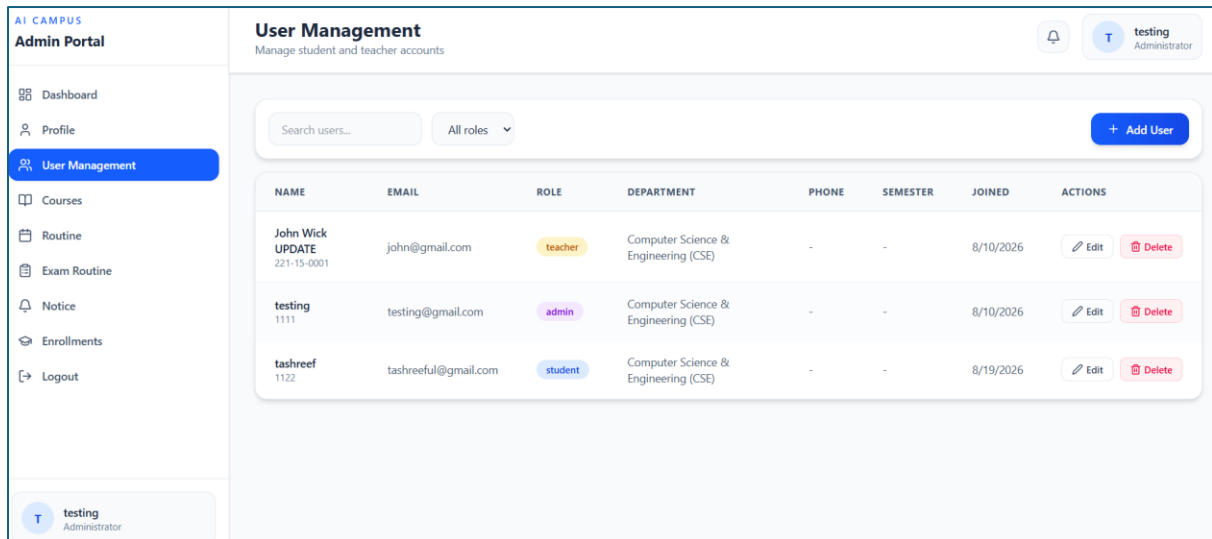


Figure 4: Admin User Management

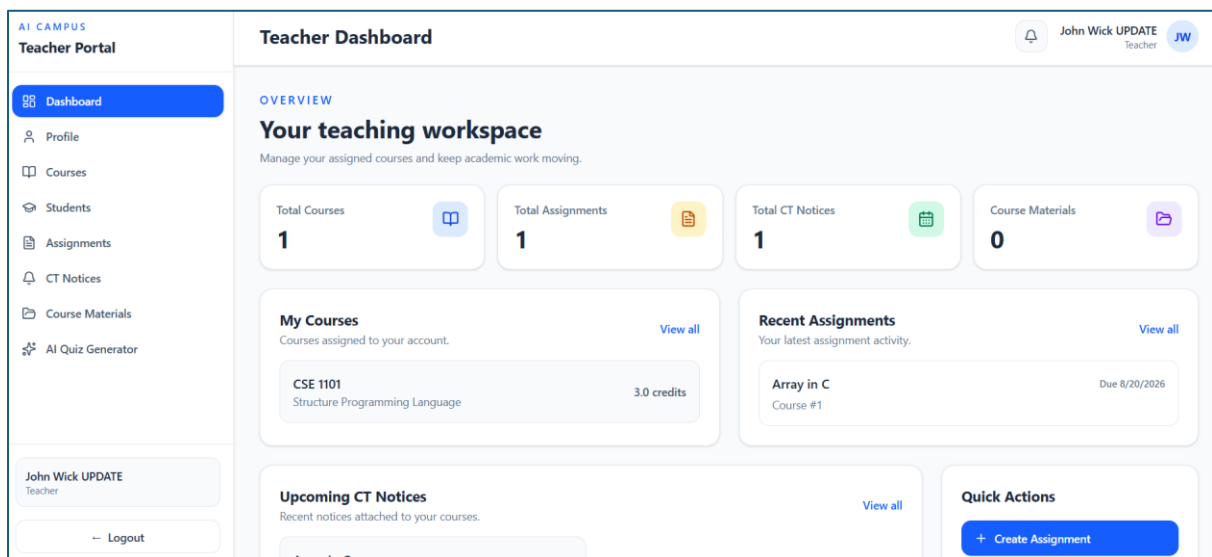


Figure 5: Teacher Dashboard

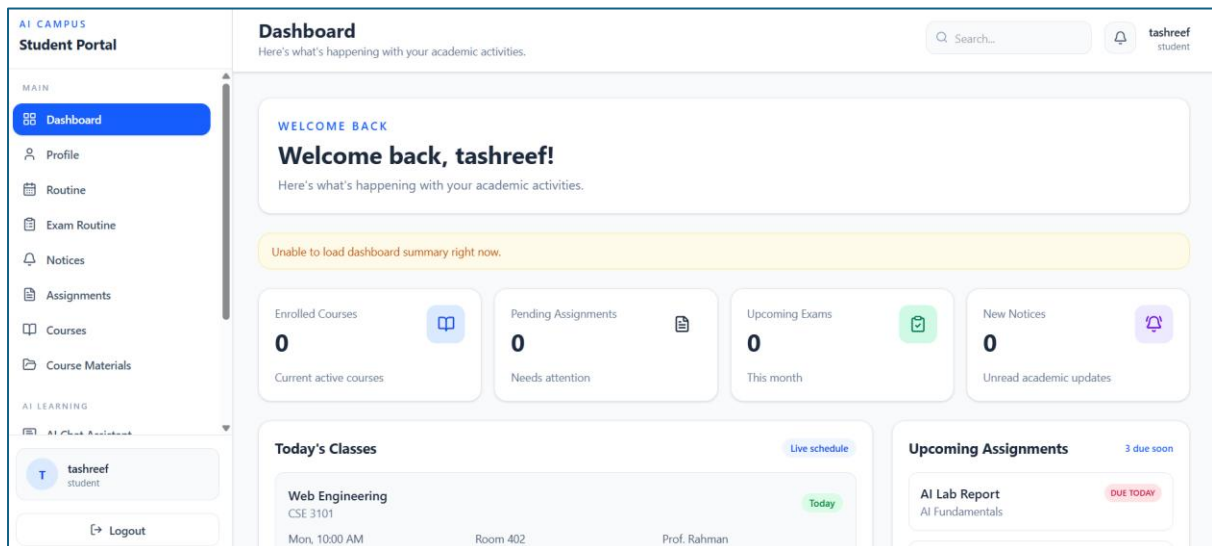


Figure 6: Student Dashboard

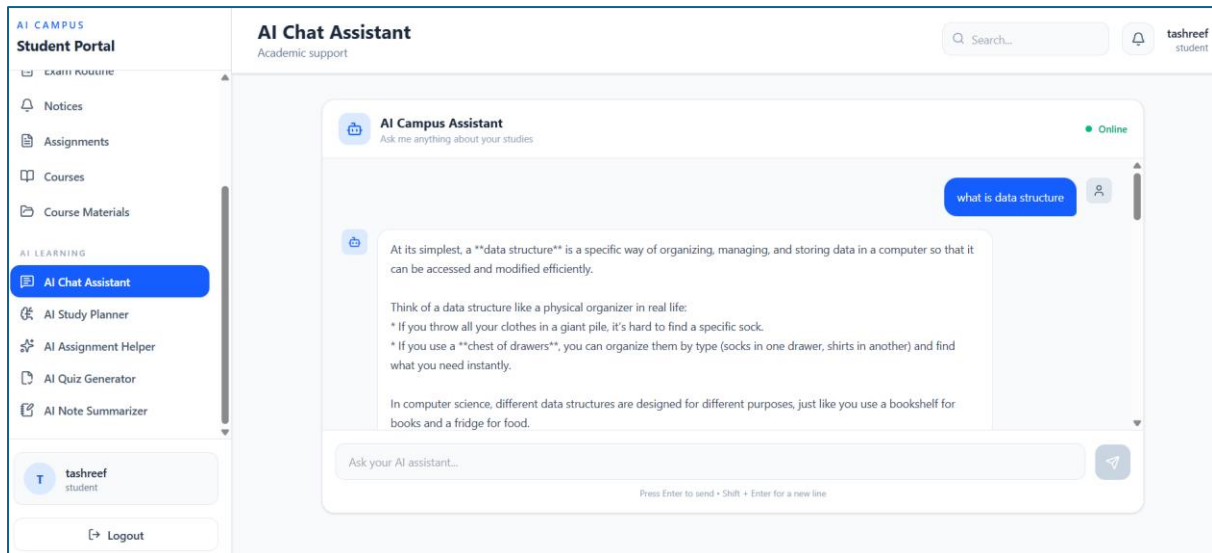


Figure 7: AI Feature-AI Chat Assistant

Table	Action	Rows	Type	Collation	Size	Overhead
ai_requests	Browse Structure Search Insert Empty Drop	23	InnoDB	utf8mb4_unicode_ci	176.0 KiB	-
assignments	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
cache	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
cache_locks	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
courses	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
course_materials	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
ct_notices	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
enrollments	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
exam_routines	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
failed_jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
jobs	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
job_batches	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
migrations	Browse Structure Search Insert Empty Drop	13	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
notices	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
password_reset_tokens	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	16.0 KiB	-
personal_access_tokens	Browse Structure Search Insert Empty Drop	19	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
routines	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
sessions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
users	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-

Figure 8: Database Module

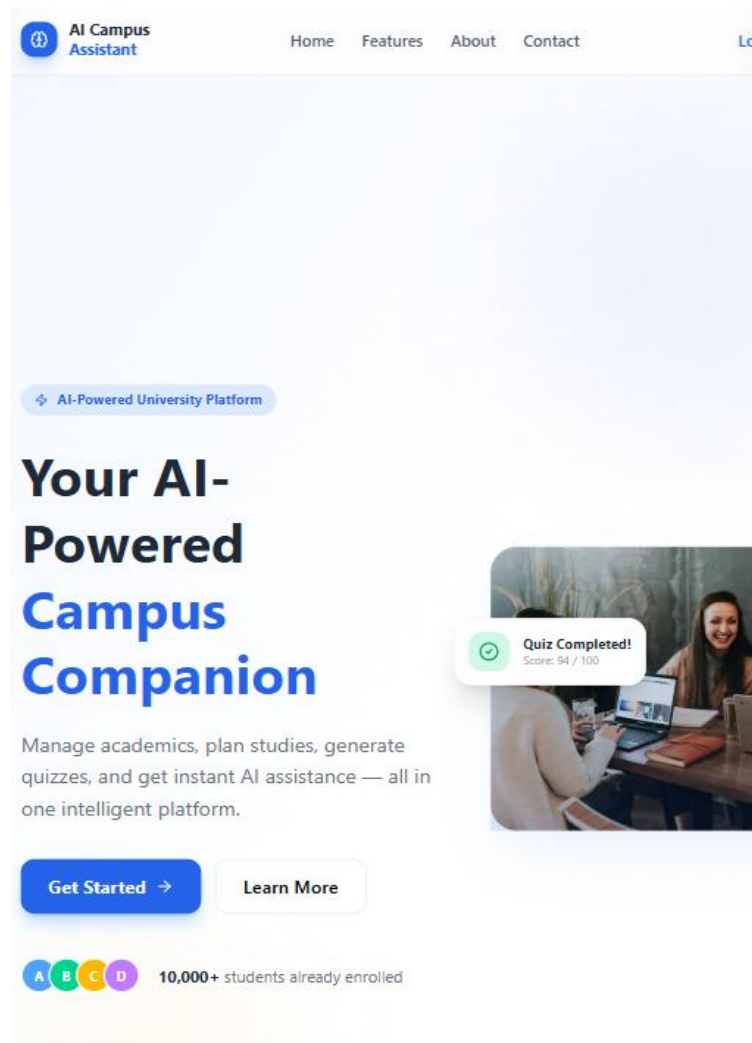


Figure 9: Responsive Interface

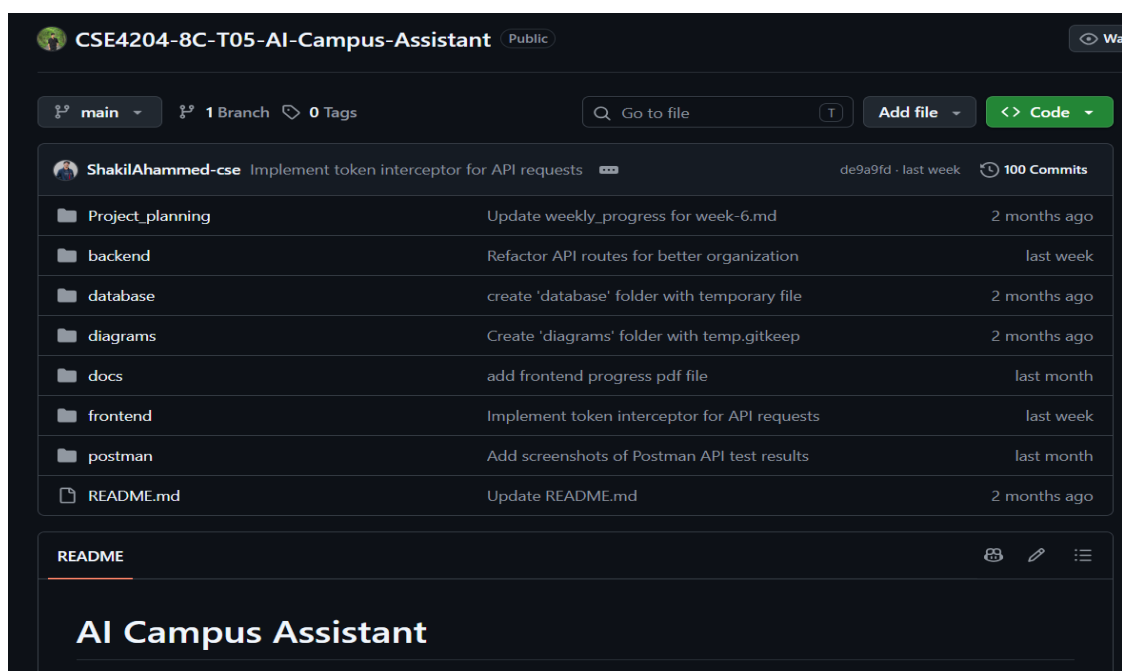


Figure 10: GitHub Repository Home

12. GitHub Repository Link

GitHub Repository:

<https://github.com/TashreefulIslam/CSE4204-8C-T05-AI-Campus-Assistant>

13. Live Demo Link

Live Demo: Not available at the moment.

Project Demonstration Video: <https://drive.google.com/drive/folders/1Mb8NLPwT6SQf-l3w95Bywb2V5olhhoaY?usp=sharing>

14. Conclusion

By Week 09, the AI Campus Assistant project has completed its major foundation and core development. The backend, database, authentication system, Admin Dashboard, Teacher Dashboard, Student Dashboard, and planned AI features have been completed.

The remaining work is mainly focused on completing the Enrollment Management and Notice Management modules in the Admin Dashboard, followed by final integration testing, debugging, and UI/UX improvements.

Overall, the project has reached the stage where the major components are working together as an integrated system and is ready to move toward the formal testing and debugging phase.